Prakruti Sudarshan | Doctoral Researcher

CONTACT INFORMATION

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NATIONALITY

Indian Citizen

EDUCATION

Ph.D. in Astronomy cum laude, IMPRS Fellow — October 2026
Max Planck Institute for Astronomy & **Heidelberg University**
Thesis: *Modeling radiation and dust dynamics in protoplanetary disks*, Advisor: Dr. Mario Flock

M.Sc. in Astro and Particle Physics, Overall Grade: 1.1 — September 2022
Eberhard Karls Universität Tübingen, Germany
Thesis: *Spirals and gravitoturbulence in 2D disks*, Advisor: Late Prof. Dr. Wilhelm Kley

Christ University, Bangalore, India — June 2020
B.Sc. *triple major* in Physics, Mathematics and Chemistry, GPA: 3.93/4.00
Research Education and Advancement Program (REAP) student in Astrophysics, Jawaharlal Nehru Planetarium, Bangalore.

RESEARCH INTERESTS

I work on **planet formation theory**, studying **dust–gas dynamics of starlight-heated protoplanetary disks** using global radiation hydrodynamical simulations with the PLUTO code and synthetic observations with radiative transfer methods. I am involved in **ongoing collaborations** modeling **circumbinary disks**, **pressureless dust fluids** and **disk instabilities**.

TECHNICAL SKILLS

Programming languages: C, Python (*Advanced*); Rust, CUDA, C++ (*Basic*)
Astrophysical Codes: PLUTO with radiation multifluid hydrodynamics, RADMC3D (*Advanced*); Rebound, IDEFIX (*Intermediate*)
High Performance Computing with MPI/GPUs (*Advanced*)
Data visualization & other tools: LaTeX, Origin Pro, Matlab, JavaScript, Go;

RESEARCH STAYS

Planet Formation Group, CCA, Flatiron Institute, New York City — Jan – Feb 2026
Visiting Student, working on a radiation-methods comparison and the dust module in PLUTO, hosted by Dr. Thomas Pfeil and Prof. Phil Armitage.

Computational Physics Group, Universität Tübingen, Germany — July 2021 – March 2022
HiWi student, worked on a project of how cooling influences circumbinary disks with the GPU–PLUTO code with Dr. Anna Penzlin, Dr. Alex Ziampras, Prof. Dr. Wilhelm Kley and Prof. Dr. Richard Nelson, related paper published in A&A (664, A157)

PUBLICATIONS: MAJOR CONTRIBUTION

Kavipurapu, P., **Sudarshan, P.**, Flock, M., et al. in prep, "Vertical dust stratification in VSI-turbulent protoplanetary disks"

Sanchez-Bermudez, J., **Sudarshan, P.**, et al., under review in MNRAS Letters,"The GRAVITY young stellar object survey: XVII. Deciphering the nature of FS CMa with infrared interferometric imaging" (GRAVITY YSO Collaboration)

Sudarshan, P., Ziampras, A., Flock, M., et al., under review in ApJS, "Multifluid dust dynamics in PLUTO: Benchmarking dust–gas drag and diffusion against existing implementations"

Baronett, S., et al. (incl. **Sudarshan, P.**), accepted in ApJ, "A Comparative Study of the Streaming Instability: Unstratified Models with Marginally Coupled Grains" [SciX]

Sudarshan, P., Flock, M., et al., 2026, A&A 706, A198 "Starlight-driven flared-staircase geometry in radiation hydrodynamics simulations of protoplanetary disks" [SciX]

Ziampras, A., **Sudarshan, P.**, et al., 2025, MNRAS, 536, 4, pp.3322-3337 "Dusty substructures induced by planets in ALMA disks: how dust growth and dynamics changes the picture" [SciX]

Sudarshan, P., Penzlin, A. B. T., Ziampras, A., Kley, W., & Nelson, R. P., 2022, A&A, 664, A157 "How cooling influences circumbinary disks" [SciX]

Co-authored Publications

Muley, D., et al. (incl. **Sudarshan, P.**), accepted in A&A, "Simulating the thermodynamics of gas, radiation, and multispecies dust: method and applications to protoplanetary disks" [SciX]

Cecil, M., et al. (incl. **Sudarshan, P.**), 2026, A&A, 707, A296 "The role of detailed gas and dust opacities in shaping the evolution of the inner disc edge subject to episodic accretion" [SciX]

Ueda, T., et al. 2024 (incl. **Sudarshan, P.**), Nature Astronomy 8, 1148-1158 (2024) "Support for fragile porous dust in a gravitationally self-regulated disk around IMLup" [SciX]

Peer Review

Astronomical Journal (part of the AAS journals) — October 2026 –

Teaching and Supervision

Summer project supervision of *Niranjan Ashok (→ Grad student at Alabama)* — Summer 2025
Topic: Temperature and radiation fields at gap edges in protoplanetary disks

Master thesis co-supervision of *Pundari Kavipurapu*, Heidelberg University — 2023 – 2025
Topic: Dust dynamics in VSI active disks with parameterized cooling laws

Advanced Astronomy Lab Tutor, Heidelberg University — Summer 2023
F36 Adaptive optics: Wavefront analysis using the Shack Hartmann Sensor

Talks and Posters

PLUTO symposium, MPIA, Heidelberg Germany (***Contributed Talk***) — June 2026
3rd PFITS+ Meeting on Hydrodynamics of Circumstellar Disks and Planet Formation (***Invited Talk***) — June 2026
Star and Planet formation seminar, European Southern Observatory (ESO), Germany — May 2026
Birnstiel Group Retreat, Fraueninsel, Chiemsee (***Invited Talk***) — April 2026
Accretion-Ejection French online seminar series (***Invited Talk***) — March 2026
SPF talk, University of Michigan, Ann Arbor — March 2026
Disks and Astrochemistry group meeting, Harvard & Smithsonian CfA, Boston — March 2026
Steward Observatory theory group meeting, Tucson, Arizona — March 2026
Hydrodynamics and Dusty Turbulence in Protoplanetary Disks, Flatiron CCA Workshop, New York City (***Invited Talk***) — February 2026
Planets and Origins group discussion, Columbia University, New York City — February 2026
Planet formation group meeting, CCA Flatiron Institute, New York City — February 2026

	ICEPF Conference 2025, Shanghai, China (***Poster***)	December 2025
	Exoplanets and Planet Formation Mini-Workshop, Tsinghua DoA, Beijing	December 2025
	Exo-ELT, European Southern Observatory (ESO), Germany (***Contributed Talk***)	November 2025
	AG Planet Formation Splinter, Görlitz, Germany (***Contributed Talk***)	September 2025
	Partially Ionized Plasmas in Astrophysics, Bergen, Norway (***Contributed Talk***)	September 2025
	Multiplicity in young stars, NBI, Copenhagen (***Contributed Talk***)	August 2025
	Planet & Star Formation Coffee, MPIA	March 2025
	Astronomical Society of India Meeting, Rourkela, India (***Contributed Talk***)	February 2025
	IMPRS retreat, Mosbach, Germany	September 2024
	PLUTO symposium, Torino, Italy (***Contributed Talk***)	September 2024
	Gravity+ conference, Grenoble, France (***Poster***)	June 2024
	Theory Group seminar, IPAG Grenoble, France	June 2024
	DisktoPlanet Group meeting, MPIA, Germany	May 2024
	Astronomy Unit Group meeting, Queen Mary University London	April 2024
	Group seminar, Imperial College London	April 2024
	MHD flows in young circumstellar disks, Ringberg, Germany (***Poster***)	September 2023
	Theory group meeting, Charles University, Prague	May 2023
	Protostars and Planets VII, Kyoto, Japan (***Poster***)	April 2023
	DisktoPlanet Group meeting, MPIA, Germany	October 2022
	The inner regions of protoplanetary disks, Ringberg, Germany (***Poster***)	September 2022
	Formation and Evolution of Planetary systems conference, Tübingen (***Contributed Talk***)	August 2022
	European Astronomical Society Annual Meeting (S22 session), Valencia, Spain (***Contributed Talk***)	July 2022

Honors and Awards

International Max Planck Research School (IMPRS-HD) Fellowship	2022 – 2026
SN Bhatt Memorial ICTS Fellowship, 1/10 selected All India *(cancelled due to Covid-19)*	2020
Christ University Physics and Electronics Club Quiz Champion	2019
Research Education and Advancement Program (REAP) scholarship	2018 – 2019

Professional Service / Outreach

SOC, "Early career astronomers and their supporters" session, European Astronomical Society Annual Meeting (EAS) Lausanne, Switzerland	July 2026
SOC, Planet formation and Exoplanets (PFE) MPIA Department Retreat, Löwenstein, Germany	May 2026
Journal Club Speaker, MPIA Girls' day outreach program	April 2026
Volunteer & Session Chair, PyConDE and PyData, Darmstadt, Germany	April 2026
Co-chair & SOC, "Early career astronomers and their supporters" session, European Astronomical Society Annual Meeting (EAS) Cork, Ireland	July 2025
Co-chair & SOC, "Early career astronomers and their supporters" session, European Astronomical Society Annual Meeting (EAS) Padova, Italy	July 2024
Disk2Planet Group mailing list organizer	2024 – 2025
LOC, Ringberg meeting on "MHD flows in young circumstellar disks"	October 2023
Volunteer, MPIA Tag der offenen Tür (Open day)	October 2023
LOC, Formation and Evolution of Planetary systems conference, Tübingen	August 2022
Mentored students from under-resourced backgrounds on STEM career pathways, Viveka Scholar Program, SVYM, Sargur Village, India	December 2023
Mediator, Virtual PSYCHE exhibition season, Science Gallery Bengaluru	April –– May 2022

OBSERVING TIME

30-inch telescope, Vainu Bappu Observatory, Kavalur, India
Time series spectra of the Sun 2 nights, 2019

INDUSTRY EXPERIENCE

Bosch Sensortec GmBH, Reutlingen, Germany April 2022 – July 2022
Working Student Intern, Worked on the analysis of MEMS-based Bosch sensor data and in-house code documentation using Python and Matlab.